

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 36 — Organic Synthesis

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Multi-Step Synthetic Routes · Retrosynthetic Analysis · Regioselectivity & Directing Effects · Protecting Groups
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED**36.1 Multi-Step Synthetic Pathway Design**

— Systematic planning of multistep routes from common precursors (benzene, methylbenzene, alkenes)

36.2 Retrosynthetic Disconnection Strategy

— Working backwards from complex target molecules (retrosynthesis); identifying disconnections

36.3 Aromatic Regioselectivity & Directing Groups— Exploiting 2,4-directing activators (–OH, –NH₂, –R) versus 3-directing deactivators (–NO₂, –COOH, –C(=O)R)**36.4 Protecting Groups & Selective Reactions**— Use of protecting groups (e.g. acylation of –NH₂ to –NHCOCH₃ to moderate reactivity and protect**High-Frequency Core Repeats (Q41–Q50)**

— The 10 most frequently tested Cambridge Paper 4 questions on Organic Synthesis from the past 10 years

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Multi-Step Synthesis of Aromatic Derivatives from Benzene — 9701/42/M/J/23/Q12 [6 Marks]

HARD · Syllabus 36.1

The synthetic pathways for the conversion of benzene into several functionalised aromatic compounds are shown in Fig. 1.1.

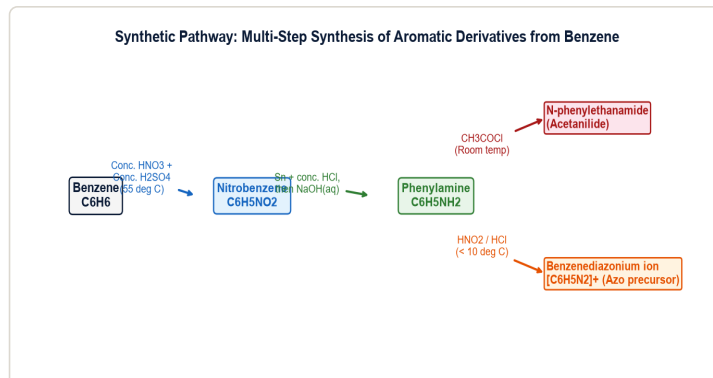


Fig. 1.1: Multi-step synthetic flowchart for producing aromatic amides and azo dye precursors from benzene.

- (a) State the reagents and conditions for the nitration of benzene to nitrobenzene. [2]
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- (b) Describe the two-stage reduction of nitrobenzene to phenylamine, stating all reagents. [2]
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- (c) State the reagent and conditions to convert phenylamine into N-phenylethanamide, and explain why this acylation moderates the reactivity of the amino group. [2]
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2 Regioselective Synthesis: 3-Bromobenzoic Acid vs 4-Bromobenzoic Acid — 9701/41/O/N/22/Q10 [6 Marks] HARD · Syllabus 36.3

A chemist needs to synthesise two different brominated benzoic acids starting from methylbenzene:

- Target 1: 3-bromobenzoic acid
- Target 2: 4-bromobenzoic acid

- (a) Devise a two-step route to synthesise Target 1 (3-bromobenzoic acid) from methylbenzene, stating all reagents, conditions, and intermediate structures. [3]
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- (b) Devise a two-step route to synthesise Target 2 (4-bromobenzoic acid) from methylbenzene, explaining why the order of steps is reversed compared to Target 1. [3]
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3 Synthesis of Benzocaine (Ethyl 4-Aminobenzoate) — 9701/42/M/J/22/Q11 [6 Marks]

HARD · Syllabus 36.1

Benzocaine is a local anaesthetic with the structure $4\text{-H}_2\text{N-C}_6\text{H}_4\text{-COOCH}_2\text{CH}_3$. It can be synthesised from 4-nitromethylbenzene via a three-step pathway.

- (a) Step 1 oxidises the methyl group of 4-nitromethylbenzene to a carboxylic acid. State the reagents and conditions. [2]

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- (b) Step 2 esterifies 4-nitrobenzoic acid with ethanol. State the reagent and catalyst for Step 2. [2]

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- (c) Step 3 reduces ethyl 4-nitrobenzoate to benzocaine. State the reducing agent and explain why this reduction is chemoselective. [2]

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4 Retrosynthetic Disconnection and Synthesis of Phenacetin — 9701/41/M/J/21/Q12 [6 Marks]

HARD · Syllabus 36.2

Phenacetin (4-ethoxyacetanilide) has the formula $4\text{-(CH}_3\text{CH}_2\text{O)-C}_6\text{H}_4\text{-NHCOCH}_3$. It is prepared starting from 4-aminophenol, $4\text{-HO-C}_6\text{H}_4\text{-NH}_2$.

- (a) State which functional group in 4-aminophenol must be reacted first, and explain why. [2]

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- (b) Write the equation for the reaction of 4-aminophenol with ethanoyl chloride to form paracetamol (4-acetamidophenol). [2]

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- (c) Describe how paracetamol is subsequently converted to phenacetin by reaction with bromoethane, stating all reagents and conditions. [2]

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5 Protecting Groups in Organic Synthesis: Amide Protection of Amines — 9701/42/O/N/20/Q10 [6 Marks]

HARD · Syllabus 36.4

When phenylamine is treated directly with concentrated nitric and sulfuric acids, tarry oxidation products are formed and extensive polysubstitution occurs.

To obtain 4-nitroaniline cleanly, a three-step protecting group strategy is used.

- (a) Step 1 protects the amino group by conversion to N-phenylethanamide. State the reagent. [1]

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- (b) Explain how the acetyl group ($-\text{COCH}_3$) protects the amino group against oxidation and prevents polysubstitution during nitration. [3]
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- (c) Step 3 removes the protecting group (deprotection). State the reagents and conditions for Step 3. [2]
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6 Multi-Step Synthesis of 2-Phenylethanoic Acid from Benzene — 9701/43/M/J/23/Q11 [6 Marks] HARD · Syllabus 36.1

Devise a three-step synthetic route to prepare 2-phenylethanoic acid, $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$, starting from benzene.

- (a) Step 1 converts benzene into methylbenzene (toluene). State the reagents, catalyst, and mechanism. [2]
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- (b) Step 2 converts methylbenzene into (chloromethyl)benzene, $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$. State the reagent and essential conditions, contrasting them with those used to chlorinate the benzene ring. [2]
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- (c) Step 3 converts (chloromethyl)benzene into 2-phenylethanoic acid via a nitrile intermediate. State the two successive reactions involved. [2]
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7 Chemoselective Reductions in Bifunctional Compounds — 9701/42/F/M/22/Q12 [6 Marks] HARD · Syllabus 36.4

Compound W has the structure $4-(\text{CH}_3\text{CO})-\text{C}_6\text{H}_4-\text{COOCH}_3$, containing both a ketone carbonyl and an ester functional group.

- (a) Predict the organic product formed when Compound W is treated with sodium tetrahydridoborate, NaBH_4 , in methanol, and explain your choice. [3]
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- (b) Predict the organic product formed when Compound W is treated with excess lithium tetrahydridoaluminate, LiAlH_4 , in dry ether, followed by dilute acid. [3]
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8 Synthesis of Azo Dyes: Regiochemistry and Extended Conjugation — 9701/41/O/N/23/Q11 [6 Marks]

HARD · Syllabus 36.1

- (a) Explain why benzenediazonium chloride couples almost exclusively at position 4 of 2-methylphenol rather than position 6. [2]

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- (b) Draw the structural formula of the azo dye produced in (a). [2]

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- (c) Explain why adding extra conjugated double bonds or aromatic rings shifts the absorption spectrum of an azo dye to longer wavelengths (bathochromic shift). [2]

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9 Synthesis of Mandelic Acid (2-Hydroxy-2-phenylethanoic Acid) — 9701/42/M/J/20/Q10 [6 Marks]

HARD · Syllabus 36.1

Mandelic acid, $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{COOH}$, is used in skin treatments.
It can be synthesised from benzaldehyde, $\text{C}_6\text{H}_5\text{CHO}$, in two steps.

- (a) Step 1 reacts benzaldehyde with hydrogen cyanide. State the reagents, catalyst, and mechanism. [3]

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- (b) Draw the structural formula of the cyanohydrin intermediate formed in Step 1. [1]

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- (c) Step 2 hydrolyses the nitrile group to a carboxylic acid. State the reagents and conditions, and explain why the mandelic acid produced is optically inactive. [2]

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10 Multi-Step Synthesis: Preparation of 1-Phenylethan-1-ol via Grignard Reagent — 9701/41/M/J/19/Q11 [6 Marks]

HARD · Syllabus 36.1

- (a) Bromobenzene is reacted with magnesium in dry ethoxyethane to form phenylmagnesium bromide. Write the equation. [1]

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- (b) Explain why completely anhydrous conditions are essential when preparing and using Grignard reagents. [2]

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- (c) Phenylmagnesium bromide is reacted with ethanal, CH_3CHO , followed by dilute acid. Draw the displayed formula of the organic product and name it. [3]

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11 Synthesis of Chiral Pharmaceuticals: Ibuprofen Synthesis Intermediates — 9701/42/O/N/21/Q11 [6 Marks] HARD · Syllabus 36.1

Ibuprofen is 2-[4-(2-methylpropyl)phenyl]propanoic acid.
One industrial route starts from 2-methylpropylbenzene (isobutylbenzene).

- (a) Step 1 reacts isobutylbenzene with ethanoyl chloride in the presence of AlCl_3 . Draw the structure of the product and name the mechanism. [2]

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- (b) Step 2 reduces the ketone carbonyl with NaBH_4 . Draw the structural formula of the alcohol formed. [2]

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- (c) Identify the chiral carbon atom in the ibuprofen molecule and explain why one enantiomer is biologically active while the other is inactive. [2]

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12 Selectivity in Side-Chain vs Ring Chlorination of Alkylbenzenes — 9701/42/M/J/18/Q11 [6 Marks] HARD · Syllabus 36.3

Ethylbenzene, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3$, can be chlorinated selectively under two different sets of conditions to produce two different monochlorinated isomers.

- (a) Condition 1 produces (1-chloroethyl)benzene, $\text{C}_6\text{H}_5\text{CH}(\text{Cl})\text{CH}_3$. State the reagents and conditions, and identify the reaction mechanism. [2]

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- (b) Explain why chlorination occurs preferentially at carbon-1 of the ethyl side-chain rather than carbon-2. [2]

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- (c) Condition 2 produces a mixture of 2-chloroethylbenzene and 4-chloroethylbenzene. State the reagents and catalyst required. [2]

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13 Synthesis of Polyamide Monomers: Adipic Acid and Hexamethylenediamine — 9701/41/O/N/18/Q11 [6 Marks] **HARD · Syllabus 36.1**

Both monomers for Nylon 6,6 can be synthesised from 1,4-dichlorobutane.

- (a) 1,4-dichlorobutane is reacted with excess potassium cyanide in ethanol under reflux. Write the equation and name the dinitrile product. [2]

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- (b) Describe how the dinitrile from (a) is converted to hexane-1,6-diamine in high yield, stating reagents and catalyst. [2]

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- (c) Describe how the dinitrile from (a) is converted to hexanedioic acid (adipic acid), stating reagents and conditions. [2]

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14 Multi-Step Synthesis: 4-Nitrobenzoic Acid to 4-Aminobenzamide — 9701/42/F/M/20/Q11 [6 Marks] **HARD · Syllabus 36.1**

Devise a three-step synthetic route to convert 4-nitrobenzoic acid, $4\text{-O}_2\text{N-C}_6\text{H}_4\text{-COOH}$, into 4-aminobenzamide, $4\text{-H}_2\text{N-C}_6\text{H}_4\text{-CONH}_2$.

- (a) Step 1 converts the carboxyl group to an acyl chloride. State the reagent. [1]

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- (b) Step 2 reacts 4-nitrobenzoyl chloride with concentrated ammonia. Write the equation and name the product. [2]

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- (c) Step 3 reduces 4-nitrobenzamide to 4-aminobenzamide. State the reducing agent and explain why this reduction does not affect the amide carbonyl. [3]

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15 Retrosynthetic Analysis of Local Anaesthetic Procaine — 9701/41/M/J/17/Q10 [6 Marks] **HARD · Syllabus 36.2**

Procaine has the structural formula: $4\text{-H}_2\text{N-C}_6\text{H}_4\text{-COO-CH}_2\text{CH}_2\text{-N(CH}_2\text{CH}_3)_2$.

- (a) Identify the ester linkage in procaine and draw the structures of the two precursor molecules formed by retrosynthetic disconnection of this ester bond. [3]

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- (b) State a reagent to activate the carboxylic acid precursor to ensure rapid coupling with the alcohol precursor. [1]

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- (c) State the reagents and conditions to convert 4-nitrotoluene into the carboxylic acid precursor. [2]

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16 Green Chemistry Principles in Organic Synthesis: Atom Economy vs E-factor — 9701/42/O/N/17/Q11 [6 Marks] HARD · Syllabus 36.1

- (a) Define the term percentage atom economy for a chemical reaction. [1]

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- (b) Compare the atom economies of preparing ethanoyl chloride from ethanoic acid using PCl_5 versus using SOCl_2 . [3]

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- (c) Explain why catalytic reactions are preferred over stoichiometric reagents in green pharmaceutical synthesis. [2]

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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Directing Groups in Electrophilic Substitution: Phenol vs Benzoic Acid — 9701/42/M/J/23/Q12(b) [4 Marks] MEDIUM · Syllabus 36.3

- (a) Explain why the $-\text{OH}$ group in phenol directs incoming electrophiles to the 2- and 4-positions. [2]

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- (b) Explain why the $-\text{COOH}$ group in benzoic acid directs incoming electrophiles to the 3-position. [2]

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18 Synthesis of Benzoic Acid from Benzene in Two Steps — 9701/41/O/N/22/Q10(b) [4 Marks] MEDIUM · Syllabus 36.1

- (a) Identify the reagents and catalyst for Step 1 to convert benzene into bromobenzene. [2]

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- (b) Describe how bromobenzene is converted into benzoic acid via a Grignard intermediate. [2]

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- 19 Selective Oxidation: Primary Alcohols vs Aromatic Rings — 9701/42/M/J/22/Q11(b) [4 Marks]** MEDIUM · Syllabus 36.4
- (a) Phenylmethanol, $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$, is oxidised to benzoic acid. State the reagents and conditions. [2]
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- (b) Explain why the benzene ring in phenylmethanol is not destroyed during this oxidation. [2]
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- 20 Synthesis of 1-Phenylethylamine from Acetophenone — 9701/41/M/J/21/Q12(b) [4 Marks]** MEDIUM · Syllabus 36.1
- (a) Phenylethanone (acetophenone), $\text{C}_6\text{H}_5\text{COCH}_3$, reacts with hydroxylamine to form an oxime. State the reagents to reduce the oxime to 1-phenylethylamine. [2]
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- (b) Draw the skeletal formula of 1-phenylethylamine and identify its chiral centre. [2]
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- 21 Regioselective Nitration: Methylbenzene vs Nitrobenzene — 9701/42/O/N/20/Q10(b) [4 Marks]** MEDIUM · Syllabus 36.3
- (a) Predict the major organic products when methylbenzene is mononitrated. [2]
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- (b) Predict the major organic product when nitrobenzene is mononitrated, and explain why the reaction temperature must be higher (90°C). [2]
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- 22 Synthesis of Phenyl Ethanoate: Choosing Between Ethanoic Acid and Ethanoyl Chloride — 9701/43/M/J/23/Q11(b) [4 Marks]** MEDIUM · Syllabus 36.1
- (a) Explain why phenol cannot be converted to phenyl ethanoate by heating with ethanoic acid in the presence of concentrated sulfuric acid. [2]
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- (b) State the reagent and conditions used to prepare phenyl ethanoate in high yield. [2]
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- 23 Friedel-Crafts Alkylation vs Acylation in Synthetic Design — 9701/42/F/M/22/Q12(b) [4 Marks]** MEDIUM · Syllabus 36.1
- (a) Give two reasons why Friedel-Crafts acylation followed by reduction (Clemmensen reduction) is preferred over direct alkylation for preparing propylbenzene. [2]
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- (b) Write the equation for the reaction of benzene with propanoyl chloride in the presence of AlCl_3 . [2]
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- 24 Synthesis of Benzenecarbonitrile and Conversion to Benzoic Acid — 9701/41/O/N/23/Q11(b) [4 Marks]** MEDIUM · Syllabus 36.1
- (a) Explain why chlorobenzene cannot be reacted with KCN to form benzenecarbonitrile (benzonitrile). [2]
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- (b) Describe how benzenediazonium chloride is converted to benzenecarbonitrile, and state how the nitrile is subsequently hydrolysed to benzoic acid. [2]
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- 25 Ester Protecting Groups for Carboxylic Acids — 9701/42/M/J/20/Q10(b) [4 Marks]** MEDIUM · Syllabus 36.4
- (a) Explain why a carboxylic acid group ($-\text{COOH}$) is often converted into a methyl ester ($-\text{COOCH}_3$) before carrying out reactions on other parts of a molecule. [2]
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- (b) State how the methyl ester protecting group is removed at the end of the synthesis to regenerate the free carboxylic acid. [2]
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- 26 Multi-Step Route to 4-Chlorobenzoic Acid from Methylbenzene — 9701/41/M/J/19/Q11(b) [4 Marks]** MEDIUM · Syllabus 36.3
- (a) Outline the two synthetic steps to prepare 4-chlorobenzoic acid starting from methylbenzene. [2]
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- (b) Explain what product would be formed if the sequence of the two steps is reversed. [2]
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- 27 Separation of Enantiomers by Chiral Resolving Agents — 9701/42/O/N/21/Q11(b) [4 Marks]** **MEDIUM · Syllabus 36.2**
- (a) Explain why enantiomers cannot be separated by fractional distillation or standard recrystallisation. **[2]**
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- (b) Explain how reaction with an enantiomerically pure chiral resolving agent allows the two enantiomers to be separated. **[2]**
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- 28 Synthesis of Phenol from Benzene via Cumene Process — 9701/42/M/J/18/Q11(b) [4 Marks]** **MEDIUM · Syllabus 36.1**
- (a) Benzene reacts with propene in the presence of an acid catalyst to form cumene (isopropylbenzene). Write the equation. **[2]**
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- (b) Cumene is oxidised by oxygen and then cleaved by dilute sulfuric acid to produce phenol and an industrially important ketone. Name this ketone. **[2]**
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- 29 Synthesis of Salicylic Acid: Kolbe-Schmitt Reaction — 9701/41/O/N/18/Q11(b) [4 Marks]** **MEDIUM · Syllabus 36.1**
- (a) Sodium phenoxide is heated under pressure with carbon dioxide to produce sodium 2-hydroxybenzoate. State the type of reaction mechanism. **[2]**
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- (b) Explain why sodium phenoxide reacts with carbon dioxide, whereas neutral phenol does not. **[2]**
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- 30 Synthesis of Aromatic Aldehydes: Gattermann-Koch Reaction vs Oxidation — 9701/42/F/M/20/Q11(b) [4 Marks]** **MEDIUM · Syllabus 36.1**
- (a) Explain why oxidising methylbenzene with hot acidified KMnO_4 does not stop at benzaldehyde. **[2]**
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- (b) State the reagent used to oxidise phenylmethanol selectively to benzaldehyde without over-oxidation to benzoic acid. **[2]**
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- 31 Synthesis of Azo Dyes: Regioselective Coupling with 1-Naphthol — 9701/41/M/J/17/Q10(b) [4 Marks]** MEDIUM · Syllabus 36.3
- (a) When benzenediazonium chloride is coupled with alkaline 1-naphthol, substitution occurs predominantly at position 4. Explain why position 4 is favoured over position 2. [2]
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- (b) Draw the skeletal formula of the resulting azo dye. [2]
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- 32 Solid-Phase Peptide Synthesis (Merrifield Method) — 9701/42/O/N/17/Q11(b) [4 Marks]** MEDIUM · Syllabus 36.4
- (a) Explain the principle of solid-phase peptide synthesis. [2]
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- (b) State one major advantage of solid-phase synthesis over traditional liquid-phase solution synthesis. [2]
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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

- 33 Classification of Directing Groups — 9701/42/M/J/23/Q12(a) [2 Marks]** EASY · Syllabus 36.3
- (a) Classify the $-\text{COCH}_3$ and $-\text{OCH}_3$ groups as 2,4-directing or 3-directing. [2]
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- 34 Role of Anhydrous AlCl_3 Catalyst in Friedel–Crafts Reactions — 9701/41/O/N/22/Q10(a) [2 Marks]** EASY · Syllabus 36.1
- (a) Explain the role of AlCl_3 in generating the electrophile from ethanoyl chloride. [2]
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- 35 Definition of Retrosynthesis — 9701/42/M/J/22/Q11(a) [2 Marks]** EASY · Syllabus 36.2
- (a) Explain what is meant by retrosynthetic analysis. [2]
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- 36 Reagent for Chemoselective Reduction of Aldehydes in Presence of Esters — 9701/41/M/J/21/Q12(a) [2 Marks]** EASY · Syllabus 36.4
- (a) Identify a reducing agent that reduces aldehydes without reducing esters, and state the solvent. [2]
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37 Directing Effect of the Nitro Group — 9701/42/O/N/20/Q10(a) [2 Marks]

EASY · Syllabus 36.3

- (a) State which ring position is substituted when nitrobenzene is brominated, and state whether nitrobenzene reacts faster or slower than benzene. [2]

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38 Byproduct of Sandmeyer Reaction — 9701/43/M/J/23/Q11(a) [2 Marks]

EASY · Syllabus 36.1

- (a) Identify the gas evolved when benzenediazonium chloride is heated with aqueous copper(I) chloride. [2]

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39 Protecting Group for Aldehydes — 9701/42/F/M/22/Q12(a) [2 Marks]

EASY · Syllabus 36.4

- (a) Name the functional group formed when an aldehyde is reacted with ethane-1,2-diol in the presence of an acid catalyst to protect the carbonyl. [2]

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40 Oxidising Agent for Side-Chain of Alkylbenzenes — 9701/41/O/N/23/Q11(a) [2 Marks]

EASY · Syllabus 36.1

- (a) Identify the reagent and condition used to oxidise an alkyl side-chain on a benzene ring directly to a -COOH group. [2]

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SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)**41 Core Repeat 1: Multi-Step Synthetic Planning from Benzene to 4-Chlorobenzoic Acid — 9701/42/M/J/22/Q10 [6 Marks]**

EASY · Syllabus 36.1

Designing correct sequences of electrophilic substitution and functional group interconversions is a hallmark of Paper 4.

- (a) Describe a three-step route to synthesise 4-chlorobenzoic acid from benzene. [3]

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- (b) Explain why carrying out the steps in a different order would fail to yield the 4-chloro isomer. [3]

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42 Core Repeat 2: Retrosynthetic Disconnection and Route Design for Paracetamol — 9701/41/O/N/21/Q11 [6 Marks]

EASY · Syllabus 36.2

- (a) Draw the structural formula of paracetamol (4-acetamidophenol) and identify the two functional groups present. [2]

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- (b) Show the retrosynthetic disconnection of the amide bond and identify the two precursor synthons. [2]

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- (c) State the forward reagents and conditions to perform this coupling reaction in the laboratory with high yield. [2]

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43 Core Repeat 3: Regiochemical Control and Protecting Groups in Aniline Nitration — 9701/42/M/J/21/Q12 [6 Marks] Syllabus 36.4

- (a) Explain what happens when phenylamine is treated directly with concentrated HNO_3 and concentrated H_2SO_4 at 55°C . [2]

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- (b) Outline the complete three-step method used to synthesise 4-nitroaniline cleanly from phenylamine. [3]

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- (c) State why this strategy successfully prevents oxidation of the nitrogen atom. [1]

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44 Core Repeat 4: Selective Oxidation and Reduction in Multi-Functional Organic Synthesis — 9701/42/O/N/19/Q11 [6 Marks] Syllabus 36.4

Compound Z is 4-(hydroxymethyl)cyclohexan-1-one.

- (a) Identify a reagent that will oxidise the primary alcohol group of Compound Z to an aldehyde without affecting the ketone group. [2]

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- (b) Identify a reagent that will reduce both the ketone and the primary alcohol (if oxidised to acid) to alcohol groups. [2]

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- (c) Explain how the ketone group could be protected as an acetal while the primary alcohol is oxidised to a carboxylic acid. [2]

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45 Core Repeat 5: Chemoselective Reduction of Unsaturated Carbonyls — 9701/42/M/J/23/Q12(c) [4 Marks] MEDIUM · Syllabus 36.4

Cinnamaldehyde has the structure $\text{C}_6\text{H}_5\text{-CH=CH-CHO}$, containing both an alkene double bond and an aldehyde group.

- (a) Identify a reducing agent that will selectively reduce the aldehyde group to an alcohol while leaving the alkene C=C intact. [2]

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- (b) Identify a reagent and catalyst that will reduce both the C=C double bond and the aldehyde group completely. [2]

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46 Core Repeat 6: Synthesis of 3-Chloromethylbenzene vs 4-Chloromethylbenzene — 9701/41/O/N/22/Q10(c) [4 Marks] MEDIUM · Syllabus 36.3

- (a) Explain why 4-chloromethylbenzene is formed as the major product when methylbenzene is chlorinated in the presence of AlCl_3 . [2]

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- (b) Describe how 3-chloromethylbenzene can be synthesised from benzene using a sequence of reactions. [2]

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47 Core Repeat 7: Synthesis of Benzyl Alcohol vs 4-Methylphenol — 9701/42/F/M/21/Q12(b) [4 Marks] MEDIUM · Syllabus 36.1

- (a) Describe a two-step route to convert methylbenzene into phenylmethanol (benzyl alcohol). [2]

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- (b) Explain why 4-methylphenol cannot be prepared by reacting 4-chloromethylbenzene with aqueous sodium hydroxide. [2]

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48 Core Repeat 8: Diazonium Coupling to Form Methyl Orange — 9701/41/M/J/20/Q11(c) [4 Marks] MEDIUM · Syllabus 36.1

- (a) State the two organic starting compounds required to synthesise the azo indicator methyl orange. [2]

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- (b) State the essential reaction conditions for the coupling step. [2]

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49 Core Repeat 9: Directing Effect of the Carbonyl Group — 9701/42/M/J/23/Q12(e) [2 Marks]

EASY · Syllabus 36.3

- (a) State which ring position is substituted when benzaldehyde, $\text{C}_6\text{H}_5\text{CHO}$, undergoes electrophilic bromination with $\text{Br}_2/\text{FeBr}_3$.

[2]

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50 Core Repeat 10: Catalyst for Nitration of Benzene — 9701/41/O/N/23/Q11(e) [2 Marks]

EASY · Syllabus 36.1

- (a) Name the acid catalyst used in the generation of the nitronium ion (NO_2^+) for the nitration of benzene.

[2]

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Mark Scheme — Topic 36 — Organic Synthesis

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Concentrated nitric acid (conc. HNO ₃) and concentrated sulfuric acid (conc. H ₂ SO ₄) [1]; Maintained at 55 °C in a water bath [1].	2
1(b)	Tin (Sn) and concentrated hydrochloric acid (conc. HCl), heated under reflux [1]; Followed by addition of excess aqueous sodium hydroxide, NaOH(aq), to liberate the free amine from its phenylammonium salt [1].	2
1(c)	Ethanoyl chloride, CH ₃ COCl, at room temperature [1]; The carbonyl group withdraws electron density from nitrogen via resonance, delocalising the nitrogen lone pair and preventing multiple unwanted substitutions or oxidation [1].	2
2(a)	Step 1: Oxidise methylbenzene with alkaline KMnO ₄ heated under reflux, followed by dilute acid, to form benzoic acid [1]; Step 2: Brominate benzoic acid with Br ₂ and FeBr ₃ catalyst [1]; The -COOH group is electron-withdrawing and directs the incoming bromine electrophile to the 3-position (meta-directing) [1].	3
2(b)	Step 1: Brominate methylbenzene with Br ₂ and FeBr ₃ in the dark to form 4-bromomethylbenzene (and 2-isomer) [1]; Step 2: Oxidise the methyl group with alkaline KMnO ₄ under reflux, followed by acidification, to form 4-bromobenzoic acid [1]; The -CH ₃ group is 2,4-directing, so bromination must precede oxidation to introduce bromine at position 4 [1].	3
3(a)	Potassium manganate(VII), KMnO ₄ , and dilute sulfuric acid (or alkaline KMnO ₄ followed by acid) [1]; Heated under reflux [1].	2
3(b)	Ethanol, CH ₃ CH ₂ OH [1]; Concentrated sulfuric acid (conc. H ₂ SO ₄) catalyst, heated under reflux [1].	2
3(c)	Tin (Sn) and concentrated HCl (or H ₂ with Pt/Ni catalyst) [1]; Chemoselective because it reduces the nitro group (-NO ₂) to an amino group (-NH ₂) without hydrolysing or reducing the ester linkage (-COOCH ₂ CH ₃) [1].	2
4(a)	The amino group (-NH ₂) [1]; Nitrogen is less electronegative than oxygen, so the -NH ₂ lone pair is more nucleophilic and attacks the acylating agent preferentially over the -OH group [1].	2
4(b)	4-HOC ₆ H ₄ NH ₂ + CH ₃ COCl → 4-HOC ₆ H ₄ NHCOCH ₃ + HCl [2].	2
4(c)	Dissolve paracetamol in aqueous sodium hydroxide, NaOH(aq), to form the phenoxide ion [1]; Add bromoethane, CH ₃ CH ₂ Br, and warm under reflux (Williamson ether synthesis) [1].	2
5(a)	Ethanoyl chloride, CH ₃ COCl (or ethanoic anhydride) [1].	1
5(b)	The carbonyl group withdraws electron density via resonance, delocalising the nitrogen lone pair into the C=O group [1]; This moderates the powerful activating effect of the -NH ₂ group, preventing uncontrolled bromination or violent oxidation by conc. HNO ₃ [1]; It directs nitration cleanly and selectively to the 4-position as a mononitro derivative [1].	3
5(c)	Boil under reflux with dilute aqueous acid (e.g. 6 mol dm ⁻³ HCl) (or hot aqueous NaOH) [1]; Followed by neutralisation with base to liberate free 4-nitroaniline [1].	2
6(a)	Chloromethane (CH ₃ Cl) and anhydrous AlCl ₃ catalyst [1]; Electrophilic aromatic substitution (Friedel-Crafts alkylation) [1].	2
6(b)	Chlorine gas (Cl ₂) and ultraviolet (UV) light / boiling under reflux (free radical substitution) [1]; Ring chlorination requires Cl ₂ with an AlCl ₃ halogen carrier in the dark at room temperature [1].	2
6(c)	Reaction with KCN in ethanol heated under reflux to form 2-phenylacetonitrile (C ₆ H ₅ CH ₂ CN) [1]; Followed by acid hydrolysis with dilute HCl (or H ₂ SO ₄) heated under reflux to yield 2-phenylethanoic acid [1].	2
7(a)	4-[CH ₃ CH(OH)]-C ₆ H ₄ -COOCH ₃ (secondary alcohol formed, ester unchanged) [2]; NaBH ₄ is a milder reducing agent that reduces aldehydes and ketones rapidly, but is too weak to reduce esters [1].	3
7(b)	4-[CH ₃ CH(OH)]-C ₆ H ₄ -CH ₂ OH and methanol (CH ₃ OH) [2]; LiAlH ₄ is a powerful reducing agent that reduces both the ketone to a secondary alcohol and the ester to a primary alcohol [1].	3
8(a)	Position 4 is activated by both the strongly 2,4-directing -OH group and the 2,4-directing -CH ₃ group [1]; Position 6 is sterically hindered by the adjacent -OH group, whereas position 4 is unhindered [1].	2
8(b)	C ₆ H ₅ -N=N-C ₆ H ₃ (2-CH ₃)(4-OH) showing azo bridge connected at C4 relative to OH [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
8(c)	Extending the delocalised π -electron system lowers the energy gap (E) between the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) [1]; Since $E = hc/\lambda$, a smaller energy gap corresponds to absorption of light with a longer wavelength λ [1].	2
9(a)	HCN and NaCN (or KCN) catalyst (or NaCN + dilute H ₂ SO ₄) [1]; Trace alkali / pH 8–9 [1]; Nucleophilic addition mechanism [1].	3
9(b)	C ₆ H ₅ CH(OH)CN (2-hydroxy-2-phenylacetonitrile) [1].	1
9(c)	Dilute hydrochloric acid, HCl(aq), heated under reflux [1]; Optically inactive because benzaldehyde has a planar carbonyl group (C=O); attack by CN ⁻ is equally probable from above or below the plane, producing an equimolar racemic mixture of enantiomers [1].	2
10(a)	C ₆ H ₅ Br + Mg C ₆ H ₅ MgBr [1].	1
10(b)	Grignard reagents are extremely strong Bronsted-Lowry bases [1]; Any moisture / water instantly protonates the carbanion to produce benzene and Mg(OH)Br, destroying the reagent: C ₆ H ₅ MgBr + H ₂ O → C ₆ H ₆ + Mg(OH)Br [1].	2
10(c)	C ₆ H ₅ -CH(OH)-CH ₃ with all bonds displayed [2]; 1-phenylethan-1-ol [1].	3
11(a)	4-(CH ₃) ₂ CHCH ₂ -C ₆ H ₄ -COCH ₃ [1]; Electrophilic aromatic substitution (Friedel-Crafts acylation) [1].	2
11(b)	4-(CH ₃) ₂ CHCH ₂ -C ₆ H ₄ -CH(OH)CH ₃ [2].	2
11(c)	Carbon-2 of the propanoic acid side-chain (-CH(CH ₃)COOH) [1]; Biological drug targets (enzymes/receptors) are chiral 3D binding sites; only one enantiomer has the correct spatial orientation of groups to fit the active site (pharmacophore) [1].	2
12(a)	Chlorine gas (Cl ₂) and ultraviolet (UV) light / boiling under reflux [1]; Free radical substitution [1].	2
12(b)	The secondary benzylic radical (C ₆ H ₅ •CHCH ₃) formed by hydrogen abstraction at C1 is stabilised by delocalisation into the adjacent benzene π -cloud [1]; This significantly lowers the activation energy compared to forming a primary radical at C2 [1].	2
12(c)	Chlorine (Cl ₂) and an anhydrous halogen carrier such as AlCl ₃ or FeCl ₃ [1]; In the dark at room temperature (electrophilic aromatic substitution) [1].	2
13(a)	Cl-(CH ₂) ₄ -Cl + 2KCN → NC-(CH ₂) ₄ -CN + 2KCl [1]; Hexanedinitrile (or 1,4-dicyanobutane) [1].	2
13(b)	Hydrogen gas (H ₂) and nickel (Ni) or platinum (Pt) catalyst at elevated temperature and pressure (or LiAlH ₄ in dry ether) [2].	2
13(c)	Dilute hydrochloric acid, HCl(aq) (or dilute H ₂ SO ₄) [1]; Heated under reflux (acid hydrolysis) [1].	2
14(a)	Thionyl chloride, SOCl ₂ (or PCl ₅) [1].	1
14(b)	4-O ₂ N-C ₆ H ₄ -COCl + 2NH ₃ → 4-O ₂ N-C ₆ H ₄ -CONH ₂ + NH ₄ Cl [1]; 4-nitrobenzamide [1].	2
14(c)	Tin (Sn) and concentrated HCl, heated under reflux, followed by aqueous NaOH (or catalytic hydrogenation with H ₂ /Pt) [2]; The nitro group is readily reduced to an amine, whereas amides are resistant to reduction under acidic catalytic hydrogenation conditions (amides require powerful LiAlH ₄ to reduce) [1].	3
15(a)	Ester linkage -COO- identified [1]; Precursor 1: 4-aminobenzoic acid, 4-H ₂ N-C ₆ H ₄ -COOH [1]; Precursor 2: 2-(diethylamino)ethanol, HO-CH ₂ CH ₂ -N(CH ₂ CH ₃) ₂ [1].	3
15(b)	Thionyl chloride, SOCl ₂ (to convert -COOH into an acyl chloride, -COCl) [1].	1
15(c)	Oxidation of methyl group with alkaline KMnO ₄ under reflux followed by acidification to give 4-nitrobenzoic acid [1]; Followed by reduction of the nitro group with Sn and concentrated HCl [1].	2
16(a)	(Molecular mass of desired product / Total molecular mass of all reactants) × 100% [1].	1
16(b)	CH ₃ COOH + PCl ₅ → CH ₃ COCl + POCl ₃ + HCl: Reactants Mr = 60 + 208.5 = 268.5; Desired Mr = 78.5; Atom economy = (78.5 / 268.5) × 100 = 29.2% [1]; CH ₃ COOH + SOCl ₂ → CH ₃ COCl + SO ₂ + HCl: Reactants Mr = 60 + 119 = 179; Atom economy = (78.5 / 179) × 100 = 43.9% [1]; SOCl ₂ gives a significantly higher atom economy and less toxic solid waste [1].	3

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
16(c)	Catalysts are regenerated and used in sub-stoichiometric amounts, vastly reducing chemical consumption [1]; Catalysts lower activation energy, enabling reactions to run at lower temperatures and pressures with reduced energy demand and fewer side products [1].	2
17(a)	The oxygen lone pair delocalises into the ring by resonance (+M effect), increasing electron density specifically at positions 2, 4, and 6 [2].	2
17(b)	The carbonyl group withdraws electron density from the ring (-M and -I effects), deactivating all positions but deactivating positions 2 and 4 far more severely than position 3; thus position 3 is the least deactivated [2].	2
18(a)	Bromine, Br ₂ (l), and iron(III) bromide, FeBr ₃ (or AlBr ₃) catalyst [2].	2
18(b)	React bromobenzene with magnesium in dry ether to form phenylmagnesium bromide [1]; Bubble in carbon dioxide gas (CO ₂) followed by acidification with dilute hydrochloric acid to liberate benzoic acid [1].	2
19(a)	Acidified potassium dichromate(VI), K ₂ Cr ₂ O ₇ / H ₂ SO ₄ (or acidified KMnO ₄) [1]; Heated under reflux [1].	2
19(b)	The delocalised aromatic π -system has immense thermodynamic stability (resonance energy ~150 kJ mol ⁻¹) [1]; The activation energy required to disrupt the aromatic ring is far higher than that needed to oxidise the benzylic C-H and C-OH bonds of the side-chain [1].	2
20(a)	Lithium tetrahydridoaluminate, LiAlH ₄ , in dry ether (or H ₂ with Ni catalyst) [2].	2
20(b)	C ₆ H ₅ -CH(NH ₂)-CH ₃ [1]; Asterisk (*) on carbon-1 showing chiral centre bonded to -H, -CH ₃ , -NH ₂ , and -C ₆ H ₅ [1].	2
21(a)	2-nitromethylbenzene (2-nitrotoluene) and 4-nitromethylbenzene (4-nitrotoluene) [2].	2
21(b)	1,3-dinitrobenzene (meta-dinitrobenzene) [1]; The first nitro group is strongly electron-withdrawing and deactivates the ring, requiring a higher temperature to overcome the higher activation energy [1].	2
22(a)	The lone pair on the phenolic oxygen is delocalised into the benzene ring, making phenol a very poor nucleophile [1]; Ethanoic acid is not sufficiently electrophilic to react with the weakly nucleophilic phenol, and equilibrium lies far to the left [1].	2
22(b)	Ethanoyl chloride, CH ₃ COCl, at room temperature [1]; In the presence of aqueous sodium hydroxide (or pyridine base) [1].	2
23(a)	Direct alkylation with 1-chloropropane leads to carbocation rearrangement giving predominantly isopropylbenzene (cumene) [1]; Direct alkylation introduces an activating alkyl group that leads to unwanted polyalkylation, whereas the acyl group deactivates the ring and stops at mono-substitution [1].	2
23(b)	C ₆ H ₆ + CH ₃ CH ₂ COCl → C ₆ H ₅ COCH ₂ CH ₃ + HCl [2].	2
24(a)	Chlorobenzene is completely inert to nucleophilic substitution because the chlorine lone pair delocalises into the benzene π -ring, giving the C-Cl bond partial double bond character [1]; Furthermore, the incoming CN ⁻ nucleophile is electrostatically repelled by the high electron density of the aromatic ring [1].	2
24(b)	Warm with copper(I) cyanide, CuCN (Sandmeyer reaction) [1]; Heat under reflux with dilute hydrochloric acid, HCl(aq) [1].	2
25(a)	The acidic proton on -COOH would react with basic or nucleophilic reagents (e.g. Grignard reagents, bases, reducing agents), quenching them [1]; Converting to an unreactive neutral ester masks the acidic proton [1].	2
25(b)	Heating under reflux with aqueous sodium hydroxide (saponification), followed by acidification with dilute hydrochloric acid [2].	2
26(a)	Step 1: Cl ₂ + AlCl ₃ in the dark to form 4-chloromethylbenzene [1]; Step 2: Alkaline KMnO ₄ under reflux followed by acidification to oxidise the -CH ₃ group to -COOH [1].	2
26(b)	If oxidation is performed first, benzoic acid is formed [1]; The -COOH group is 3-directing, so subsequent chlorination would produce 3-chlorobenzoic acid rather than the desired 4-isomer [1].	2
27(a)	Enantiomers have identical physical properties (boiling point, melting point, density, solubility) in an achiral environment [2].	2
27(b)	Reaction with a single enantiomer of a chiral acid/base converts the enantiomers into a pair of diastereomers [1]; Diastereomers have different physical properties (different solubilities and boiling points) and can be separated by standard fractional crystallisation [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
28(a)	$C_6H_6 + CH_3CH=CH_2 \rightarrow C_6H_5CH(CH_3)_2$ [2].	2
28(b)	Propanone (acetone), CH_3COCH_3 [2].	2
29(a)	Electrophilic aromatic substitution (Kolbe-Schmitt carboxylation) [2].	2
29(b)	Carbon dioxide is a very weak electrophile [1]; The phenoxide ion carries a full negative charge and is vastly more electron-rich and activated towards electrophilic attack than neutral phenol [1].	2
30(a)	$KMnO_4$ is a powerful oxidising agent; the aldehydic C-H bond in benzaldehyde is oxidised far more rapidly than the initial methyl group, so oxidation proceeds directly to benzoic acid [2].	2
30(b)	Pyridinium chlorochromate (PCC) in anhydrous dichloromethane (or warm acidified $K_2Cr_2O_7$ with immediate distillation of the volatile aldehyde) [2].	2
31(a)	Position 4 is strongly activated by the -OH group and experiences less steric hindrance than position 2, which is adjacent to the bulky peri-hydrogen at position 8 and the -OH group [2].	2
31(b)	Structure showing 1-naphthol with azo group (-N=N-C ₆ H ₅) attached at position 4 [2].	2
32(a)	The C-terminal amino acid is covalently anchored to an insoluble polymer resin bead; successive N-protected amino acids are added step-by-step to extend the chain [2].	2
32(b)	Excess reagents and soluble byproducts can be washed away simply by filtration without time-consuming isolation and purification after every step; the process is easily automated [2].	2
33(a)	-COCH ₃ is 3-directing (meta-directing) [1]; -OCH ₃ is 2,4-directing (ortho/para-directing) [1].	2
34(a)	$AlCl_3$ acts as a Lewis acid and accepts a lone pair of electrons from chlorine [1]; Generating the powerful acylium ion electrophile: $CH_3COCl + AlCl_3 \rightarrow CH_3CO^+ + AlCl_4^-$ [1].	2
35(a)	A technique for planning an organic synthesis by breaking down the target molecule into simpler precursor structures (synthons) [1]; Working backwards step-by-step until readily available starting materials are reached [1].	2
36(a)	Sodium tetrahydridoborate, $NaBH_4$ [1]; In methanol (or ethanol / water) [1].	2
37(a)	Position 3 (meta position) [1]; Nitrobenzene reacts much slower than benzene (the -NO ₂ group is deactivating) [1].	2
38(a)	Nitrogen gas, N_2 [2].	2
39(a)	Acetal (or cyclic acetal / 1,3-dioxolane) [2].	2
40(a)	Alkaline potassium manganate(VII), $KMnO_4$ / OH ⁻ (followed by dilute acid) [1]; Heated under reflux [1].	2
41(a)	Step 1: Methylation of benzene with CH_3Cl and anhydrous $AlCl_3$ to form methylbenzene [1]; Step 2: Chlorination of methylbenzene with Cl_2 and $AlCl_3$ in the dark to yield 4-chloromethylbenzene (and 2-isomer) [1]; Step 3: Reflux with alkaline $KMnO_4$ followed by dilute acid to oxidise the methyl group to 4-chlorobenzoic acid [1].	3
41(b)	The methyl group is 2,4-directing, directing chlorination to positions 2 and 4 [1]; If methylbenzene were oxidised to benzoic acid first, the resulting -COOH group is strongly 3-directing [1]; Chlorination of benzoic acid would yield 3-chlorobenzoic acid instead of the desired 4-chloro isomer [1].	3
42(a)	4-HOC ₆ H ₄ NHCOCH ₃ [1]; Phenol group (-OH) and secondary amide group (-NHCO-) [1].	2
42(b)	Disconnection across C-N bond of -NH-CO- [1]; Synthons / precursors: 4-aminophenol (4-HOC ₆ H ₄ NH ₂) and an acetylating agent (ethanoic anhydride or ethanoyl chloride) [1].	2
42(c)	Ethanoic anhydride, $(CH_3CO)_2O$ (or CH_3COCl) [1]; Warm gently with stirring; ethanoic anhydride avoids toxic HCl fumes and gives high purity [1].	2
43(a)	Concentrated acid protonates the amino group to form the phenylammonium ion ($C_6H_5NH_3^+$), which is strongly 3-directing rather than 4-directing [1]; Strong oxidising conditions destroy the activated amine ring, forming dark tarry oxidation products [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
43(b)	Step 1: Acylate phenylamine with CH_3COCl to form N-phenylethanamide (acetanilide) [1]; Step 2: Nitrate with conc. HNO_3 + conc. H_2SO_4 at 55°C to introduce $-\text{NO}_2$ cleanly at position 4 [1]; Step 3: Hydrolyse with boiling dilute HCl followed by NaOH to remove the acetyl group and regenerate 4-nitroaniline [1].	3
43(c)	The acetyl group delocalises the nitrogen lone pair into the carbonyl π -bond, protecting the nitrogen atom against oxidation and preventing protonation [1].	1
44(a)	Pyridinium chlorochromate (PCC) in anhydrous dichloromethane [2].	2
44(b)	Lithium tetrahydridoaluminate, LiAlH_4 , in dry ether followed by dilute acid [2].	2
44(c)	React with ethane-1,2-diol and trace acid catalyst to convert ketone into a stable cyclic acetal [1]; Oxidise the primary alcohol with KMnO_4/H^+ , then hydrolyse the acetal with dilute aqueous acid to restore the ketone [1].	2
45(a)	Sodium tetrahydridoborate, NaBH_4 (in methanol or ethanol) [2].	2
45(b)	Hydrogen gas (H_2) and nickel (Ni) or platinum (Pt) catalyst [2].	2
46(a)	The methyl group is an electron-donating (+I) alkyl group that activates the ring and directs incoming electrophiles to the 2- and 4-positions [2].	2
46(b)	Chlorinate benzene to chlorobenzene ($\text{Cl}_2/\text{AlCl}_3$), then carry out Friedel-Crafts alkylation, or chlorinate benzaldehyde at position 3 and reduce the carbonyl to $-\text{CH}_3$ via Clemmensen reduction [2].	2
47(a)	Step 1: Cl_2 with UV light / reflux to form (chloromethyl)benzene, $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ [1]; Step 2: Warm with aqueous sodium hydroxide, NaOH(aq) , to hydrolyse to $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ [1].	2
47(b)	In 4-chloromethylbenzene, the chlorine atom is directly bonded to the aromatic ring [1]; The C-Cl bond has partial double bond character due to delocalisation and resists nucleophilic substitution by OH^- [1].	2
48(a)	Diazotised 4-aminobenzenesulfonic acid (sulfanilic acid) [1]; N,N-dimethylphenylamine (N,N-dimethylaniline) [1].	2
48(b)	Temperature maintained strictly between 0°C and 10°C [1]; Weakly acidic to neutral aqueous medium [1].	2
49(a)	Position 3 (meta position) [2].	2
50(a)	Concentrated sulfuric acid, conc. H_2SO_4 [2].	2